

Student ID:

Lab 09 Task: Building ChatIUT

You are tasked to build a simple multi-user chat application called **ChatIUT**.

Task Requirements

1. `tcp_chat_server.py`

- Bind to `0.0.0.0` on port `6006` (or a CLI-configured port).
- Accept multiple TCP client connections and maintain a list of active clients.
- For each incoming message from a client, forward it (broadcast) to all other connected clients.
- Handle client disconnects and remove the client from the list to avoid broken pipes.
- Use `threading.Thread(..., daemon=True)` to serve each client concurrently.
- Log server events (startup, connect, disconnect, forwarded messages) to `server.log`.

2. `tcp_chat_client.py`

- Accept server IP and port as command-line arguments: `python3 tcp_chat_client.py <server_ip> <port>`.
- Connect to the server and spawn two threads:
 - Read from `stdin` and send messages to the server.
 - Receive messages from the server and display them to the user.
- Allow a client to cleanly exit (`/quit` or `Ctrl+C`).

3. Optional (bonus) features

- Support `/nick <name>` so clients can set display names.
- Implement a UDP-based presence announcer (simple periodic broadcast).
- Allow private messages using a `/msg <client-id> <text>` command.

4. Deliverables

- (a) Source files: `tcp_chat_server.py`, `tcp_chat_client.py`, (optional `udp_discover.py`), and `README.md`.
- (b) `server.log` capturing a short session.
- (c) Short report and a couple of screenshots or terminal logs demonstrating multi-client chat.

Lab 09 Grading Criteria

1. Functionality (30 pts)

- Server accepts multiple clients and broadcasts correctly: 10 pts
- Client sends and receives messages correctly: 10 pts

- Graceful disconnect handling: 10 pts
2. Robustness & Logging (10 pts)
 - Informative logging present and correct: 5 pts
 - Error handling, retries/timeouts where applicable: 5 pts
 3. Documentation & Testing (10 pts)
 - README, run instructions, logs/screenshots: 5 pts
 - Demonstration during lab or recorded evidence: 5 pts
 4. Bonus features (30 pts)
 5. **Total: 50 pts**